

Felipe Valenzuela Gaete

Date of birth: January 12, 1988

Address: Av. Ejército 441
School of Industrial Engineering
Universidad Diego Portales
Santiago Centro, Santiago 8370191, Chile

Phone: +569-6-647-2796

Email: f.valenzuela.gaete@gmail.com

Website: <http://www.felipevalenzuela.cl>

Education

Universidad de Santiago de Chile: 08/2012 - 12/2013

Master of Science in Mechanical Engineering, *summa cum laude*, GPA: 6.9/7.0

Thesis: "Numerical Investigation of Flow Over an Airfoil by the Lattice Boltzmann Method using Multi-Block Methods"

Universidad de Santiago de Chile: 03/2008 - 12/2013

Bachelor of Science in Mechanical Engineering, *summa cum laude*, GPA: 6.9/7.0

Focus: Fluid Mechanics

Liceo Industrial Alberto Widmer: 03/2002 - 12/2005

Aircraft Maintenance Technician

Professional Experience

Universidad Diego Portales,

School of Industrial Engineering: 08/2014 - To date

Research Assistant

Accelerating a numerical code based on the Lattice Boltzmann programming massively parallel processors of a CPU by means of CUDA compute platform. Project: Fondecyt Iniciación 11121553 P.I: Andrés Díaz (andres.diaz@udp.cl)

Research Assistant

Developed a two-phase fluid dynamics code to simulate the fall of a drop on a hot surface. Project: Fondecyt Iniciación 11121553 P.I: Andrés Díaz (andres.diaz@udp.cl)

Alter Energy SpA: 01/2015 - 04/2015

Research Engineer

Developing the hydrodynamic design of an undershot water wheel used to generate electric power in small channels. Eco-Boya CORFO Project, Etapa I.

SimuLab Ltda.: 06/2014 - 12/2014

Research Engineer

Developing technical reports about the operation of a code based on the Discrete Element Method for mineral milling simulations. CORFO Project 14-PIE - 26935.

*Universidad Diego Portales,
School of Industrial Engineering: 01/2013 - 08/2014*

Research Assistant

Developed an improved refinement algorithm based on the multi-block method for Lattice Boltzmann incompressible viscous flow simulations. This scheme allows increase the computational efficiency by decreasing the CPU time compared to the conventional multi-block method. Project: Fondecyt Iniciación 11121553 P.I: Andrés Díaz (andres.diaz@udp.cl)

Research Assistant

Implemented a high order boundary treatment to the research group code. This implementation allows to accurately simulate complex geometries with the Lattice Boltzmann square mesh. Project: Fondecyt Iniciación 11121553 P.I: Andrés Díaz (andres.diaz@udp.cl)

Research Assistant Developed a numerical code based on the Lattice Boltzmann method implementing Non-uniform meshes (Multi-Block Method). This code allows solving fluid dynamic problems improving the results accuracy compared to conventional Lattice Boltzmann codes. Project: Fondecyt Iniciación 11121553 P.I: Andrés Díaz (andres.diaz@udp.cl)

*U. de Santiago de Chile,
Department of Mechanical Engineering: 08/2012 - 12/2012*

Research Assistant

Developed a numerical code based on the Lattice Boltzmann method in uniform meshes. This code allows to solve fluid dynamic benchmark problems.

Teaching Experience

Preuniversitario Nacional: 03/2010 - To date

Lecturer

Mathematics and Physics Courses

These courses prepare students for getting accepted into the university.

Universidad Diego Portales: 03/2013 - To date

Teaching Assistant

Fluid Mechanics

Teaching assistant and guest lecturer in undergraduate level courses.

Universidad de Santiago de Chile: 03/2012 - 12/2012

Teaching Assistant

Fluid Mechanics II

Teaching assistant and guest lecturer in undergraduate level courses.

Universidad de Santiago de Chile: 03/2012 - 07/2012

Teaching Assistant

Computational Code Analysis

Teaching assistant and guest lecturer in undergraduate level course, proposing engineering problems and giving numerical solutions by using both Matlab codes and Fortran codes.

Research Interests

Computational fluid dynamics and heat transfer, microfluidics heat transfer, Parallel computing, High resolution schemes, Lattice Boltzmann Method, Mesh refinement schemes, Adaptive meshing.

Awards and Honors

2014 Graduation Rank: 1st of 24th.

2014 M.S. graduation *summa cum laude*.

2014 B.S. graduation *summa cum laude*.

2013-2014 Graduate Academic Excellence Armando Quezada Fellowship.

2012-2014 Universidad de Santiago de Chile Master of Sc. in Engineering Fellowship.

2008-2013 National Fellowship for undergraduate programs.

Publication

Journal Papers

Valenzuela, F. A., Díaz, A. J., Guzmán, A. M. & Escobar, R. (2015). Efficient multi-block scheme for the Lattice Boltzmann Method. Under consideration for publication in Journal of Computational Physics

Conference Papers

Valenzuela, F. A., Guzmán, A. M., & Díaz, A. J. (2013, Nov.). Numerical Investigation of Flow Over an Airfoil by the Lattice Boltzmann Method. In ASME 2013 International Mechanical Engineering Congress and Exposition (pp. V001T01A002-V001T01A002). American Society of Mechanical Engineers. doi:10.1115/IMECE2013-62386

Master Thesis

“Numerical Investigation of Flow Over an Airfoil by the Lattice Boltzmann Method using Multi-Block Methods”, M.S. Thesis, Universidad de Santiago de Chile, Santiago, 2014.

Conference Presentations

Valenzuela, F. A., Guzmán, A. M., & Díaz, A. J. Numerical Investigation of Flow Over an Airfoil by the Lattice Boltzmann Method. In ASME 2013 International Mechanical Engineering Congress and Exposition. San Diego, California, November 15-21, 2013.

Valenzuela, F. A., Díaz, A. J., Guzmán, A. M., & García, C. Optimization of the airfoil study by means of Lattice Boltzmann with non-uniform mesh grids. In Chilean Society for Computational Mechanics (CSCM) Journey of Computational Mechanics. Santiago, Chile, October 3-4, 2013

Languages / Software

Lenguages

Fluent english and spanish

Software

Fortran and C++ programming, CUDA massively parallel programing, OpenMP parallel programming, ANSYS Fluent, Solid Works, Auto CAD, Mathcad, Matlab, Tecplot.

References

Andrés Díaz Alarcón, Ph.D.
Assistant Professor
Universidad Diego Portales
andres.diaz@udp.cl

Amador Guzmán Cuevas, Ph.D.
Associate Professor
Pontificia Universidad Católica de Chile
aguzman@ing.puc.cl

Williams Calderón, Ph.D.
Assistant Professor
Universidad de Chile
wicalder@ing.uchile.cl